

BMJ Open ADDRESS-LC: study protocol for a phase 2, triple-blind, randomised, placebo-controlled trial to evaluate the efficacy and safety of bezisterim (NE3107) in adults with long COVID

Penelope Markham ¹, Michael J Peluso,² Sherry Hsiang-Yi Chou,³ Lisa McCorkell,⁴ Stephen O'Quinn,⁵ Christopher Reading,¹ Clarence Ahlem,¹ Chantal M Petit,⁶ Jiayan Yan,¹ Joseph M Palumbo¹

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For numbered affiliations see end of article.

Correspondence to

Dr Penelope Markham;
pmarkham@bioviepharma.com

ABSTRACT

Introduction Long COVID is a debilitating condition consisting of prolonged neurological symptoms such as cognitive impairment ('brain fog') and sleep disturbance. Symptoms may be associated with ongoing neuroinflammation from the persistence of a viral reservoir, activation of other viruses, protracted macrophagic inflammatory memory or enduring viral Spike protein that stimulates proinflammatory cytokines via toll-like receptor (TLR) signalling. Bezisterim (NE3107) is an oral, blood-brain barrier-permeable, anti-inflammatory, insulin-sensitising dehydroepiandrosterone derivative that inhibits TLR-driven neuroinflammation and is being developed for neurodegenerative diseases in which TLR-driven inflammation contributes to cognitive decline.

Methods and analysis ADDRESS-LC (NCT06847191) is a phase 2, triple-blind, placebo-controlled, randomised proof-of-concept study to evaluate the efficacy and safety of bezisterim in adults with long COVID. This multicentre study will recruit 208 adults (aged 18–69 years) with symptoms of fatigue and neurocognitive impairment for at least 3 months following an index SARS-CoV-2 infection. Individuals who meet all eligibility criteria will be randomised 1:1 to receive bezisterim 20 mg or placebo two times per day for 12 weeks. Cognition, fatigue, sleep, post-exertional malaise, quality of life and burdensome symptoms will be evaluated via several assessments, including a bespoke Cogstate Cognition Battery, Patient-Reported Outcomes Measurement Information System (PROMIS) Cognitive Function Short Form (SF)-8a, PROMIS Fatigue SF-13a and Patient's Global Impression of Severity, among others.

Ethics and dissemination The protocol has been approved by an institutional review board prior to study initiation. The study is being conducted at multiple clinical research sites across the USA. A current list of participating study sites is publicly available through the ClinicalTrials.gov registration record (NCT06847191). The study protocol has received central institutional review board approval (Advarra #Pro00081088), and participating sites obtained the necessary approvals and agreements required for study conduct and data access in

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Randomised, placebo-controlled, multicentre design enhances internal validity.
- ⇒ Triple-masked design minimises bias across participants and investigators.
- ⇒ Eligibility confirmed by sponsor Enrollment Authorization Committee.
- ⇒ Short post-treatment follow-up limits assessment of durability of response.
- ⇒ Absence of validated biomarkers for long COVID introduces diagnostic heterogeneity.

accordance with applicable regulations and Good Clinical Practice guidelines. Authorised representatives of the sponsor, regulatory authorities and ethics committees may access study-related records for monitoring, auditing and regulatory purposes consistent with participant informed consent. The study will disseminate results through academic publication and conference presentations.

Trial registration number [NCT06847191](https://clinicaltrials.gov/ct2/show/study/NCT06847191).

INTRODUCTION

Post-acute sequelae of COVID-2019 (PASC/COVID-19), often referred to as long COVID, is an infection-associated chronic condition that occurs after SARS-CoV-2 infection and is present for at least 3 months as a continuous, relapsing and remitting, or progressive disease state that affects one or more organ systems.^{1–3} While there may be some protective effects from vaccination, long COVID was still estimated to affect ~400 million individuals worldwide in 2023, and data collected from 2022 to 2024 via the US Census Bureau Household Pulse Survey estimate that ~18% of adults in the USA have experienced long COVID.^{4,5}

Individuals with long COVID may experience a broad range of mild to severe symptoms that can persist for months or years following the inciting infection.^{1 2} Long COVID remains a major public health concern, affecting millions of US adults, with the associated symptoms leading to significant decreases in quality of life and potentially long-term disability.^{3 6} Fatigue and brain fog (ie, cognitive impairment manifesting as poor memory or difficulty in thinking or concentrating) are among the most debilitating symptoms and can have a profound impact on daily life and the ability to work.^{7–10} Regardless of type or severity, neurological symptoms following a SARS-CoV-2 infection can be relatively long lasting, with one study showing 85% of patients experiencing continued long COVID symptoms 12 months after symptom onset.^{11 12}

Long COVID has also been associated with an increased risk for myalgic encephalomyelitis/chronic fatigue syndrome and postural orthostatic tachycardia syndrome (POTS).^{13 14} These conditions, which also include deficits in memory, attention, executive function and processing speed, can be triggered by viral infections.^{13–17} Shared clinical features between infection-associated chronic conditions suggest that there may be similar underlying pathology.^{15 16 18 19}

Mechanisms underlying the neurological symptoms of long COVID require further characterisation, but chronic inflammation is thought to be a significant contributor.^{20 21} Growing evidence suggests that cognitive impairments are related to a chronic inflammatory state, potentially caused by persistence of a viral reservoir, activation of other viruses and/or protracted macrophagic inflammatory memory.⁷ Specifically, long COVID has been associated with persistent plasma elevations of proinflammatory cytokines that modulate neurocircuitry related to learning and memory under normal physiologic conditions, but that may promote neuronal death and cognitive dysfunction when their equilibrium shifts to a chronic, elevated state.^{3 22} Additionally, COVID-19 can trigger epigenetic changes that affect post-infection equilibrium among these cytokines.²³ Blood–brain barrier (BBB) disruption can occur during both acute infection and long COVID, and the extent of this disruption is strongly linked to cognitive impairment.²⁴ Recent positron emission tomography data demonstrated increased glial activation in long COVID, further supporting the role of a neuroimmune response in neurological symptoms.^{25 26}

SARS-CoV-2 Spike protein is a highly inflammatory protein that mediates cell entry and has been shown to cross the BBB.²⁷ Spike protein has been detected in skull, meninges and brain samples from patients who died from non-COVID-19-related causes long after infection.²⁸ Additionally, circulating SARS-CoV-2 Spike protein has been detected in patients with long COVID for up to 14 months post infection, suggesting a role for persistent Spike protein in long COVID symptoms.^{29–31} One postulated mechanism is that Spike protein binds to membrane receptor for advanced glycation end products and toll-like

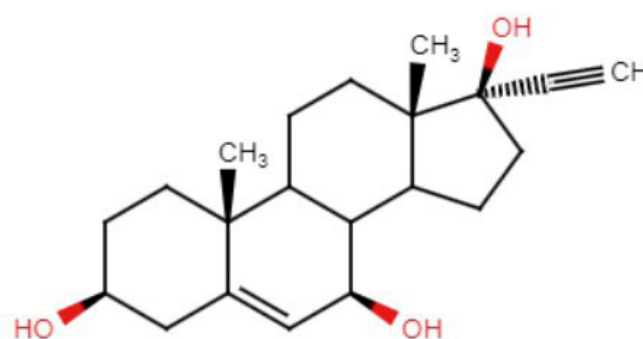


Figure 1 Bezisterim, 17 α -ethynyl-androst-5-ene-3 β , 7 β -triol.

receptor 4 (TLR4) with high specificity, creating a proinflammatory feedback loop involving the interleukin-6 (IL-6), tumour necrosis factor (TNF) and IL-1 β triad.^{32 33} Specifically, Spike S1 protein may activate TLR4, subsequently promoting proinflammatory cytokine expression via nuclear factor kappa B (NF- κ B) and extracellular signal-regulated kinase (ERK) signalling.^{33–37} This excess production of inflammatory mediators could serve as a mechanism for neurological impairment given the functional role of these cytokines in mood and cognition.²¹ Consistent with this, rodent studies have shown that Spike protein induces cognitive deficits and neurodegeneration via TLR4 activation.³⁴ A recent study found that hippocampal injection of S1 protein in mice produced cognitive impairment, neuronal loss and neuroinflammation, with the corresponding inflammatory phenotype significantly reduced by pretreatment with a selective TLR4 inhibitor.³⁶ Additionally, TLR4 variants have been associated with cognitive outcomes and severity in long COVID.^{19 34}

Bezisterim (NE3107) is a synthetic analogue of a pharmacologically active metabolite of dehydroepiandrosterone, androst-5-ene-3 β -7 β -17 β -triol (figure 1), that has anti-inflammatory and insulin-sensitising activity, without being immunosuppressive, as well as good oral bioavailability and BBB permeability.^{38 39} Based on current studies, bezisterim's anti-inflammatory mechanisms may have the potential to reduce the neurocognitive symptoms associated with long COVID by inhibiting TLR-driven phosphorylation of ERK and NF- κ B signalling.^{38–47} In vitro studies have demonstrated that bezisterim binds ERK and reduces proinflammatory signalling downstream of TLR4 activation leading to decreased activation and nuclear translocation of NF- κ B and the expression of proinflammatory cytokines.^{39 42 43} Preclinical studies have demonstrated BBB permeability, antineuroinflammatory and neuroprotective properties in rodent models of neurodegeneration, including Parkinson's disease (PD), through reduced activation of NF- κ B, reduced microglial activity, improved motor function and reduced inflammation.^{45–47} In exploratory analyses of a subgroup of 50 subjects with mild to moderate probable Alzheimer's disease, improvements in cognitive and functional performance measures showed correlations with biological age (determined by

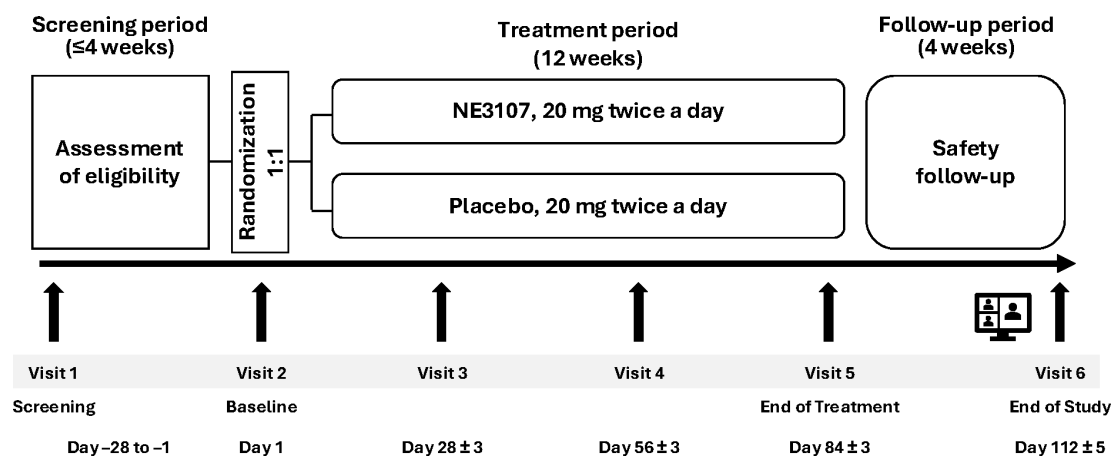


Figure 2 Study design. mg, milligramme; NE3107, bezisterim.

DNA methylation ‘clocks’), metabolism, inflammation and dementia biomarkers in those treated with bezisterim compared with placebo.³⁸ Bezisterim was well tolerated with no potential safety signals identified. In vitro studies indicate that bezisterim is primarily metabolised by cytochrome P450 (CYP)3A4 without evidence of inhibition or induction, and metabolites found in human plasma have not shown CYP3A4 and CYP2C9 inhibition, suggesting a low potential for drug:drug interactions. Collectively, the mechanism of action and the safety data for bezisterim indicate that it is likely to have minimal safety risks and may provide a beneficial effect on the cognitive symptoms of long COVID.⁴⁴

METHODS AND ANALYSIS

Study design

ADDRESS-LC is an ongoing phase 2, randomised, placebo-controlled, proof-of-concept, multicentre trial to evaluate the safety and efficacy of bezisterim in US adults 18–69 years of age with long COVID symptoms lasting ≥ 3 months (NCT06847191). This study uses a triple-masked design in which participants, investigators, outcome assessors and sponsor personnel remain blinded to treatment allocation throughout the study. The trial design includes a screening period of up to 4 weeks, a 12-week intervention period and a 4-week follow-up period, for a total duration of up to 20 weeks (figure 2). There are six planned visits during which written informed consent, medical history, labs and administration of clinical outcome assessments, such as the bespoke Cogstate Cognition Battery, various patient-reported outcome (PRO) assessment tools, Patient Health Questionnaire-2 (PHQ-2) and Columbia-Suicide Severity Rating Scale (C-SSRS) Screening/Base-line Form will be obtained. See online supplemental exhibit 1 for full details and Schedule of Activities. The total time required to complete all assessments at site visits during the treatment period is approximately 2 hours, although the overall visit duration may be longer due to optional breaks between assessments. Assessments at the screening visit, which include additional PRO assessments for eligibility, take approximately 3 hours to complete.

Eligibility is confirmed by the sponsor Eligibility Authorization Committee. During the baseline (BL) visit (visit 2), individuals meeting the eligibility criteria are stratified by age (≥ 18 to ≤ 45 years or > 45 to < 70 years) and symptom duration (< 2 years or ≥ 2 years) and randomised 1:1 to receive either bezisterim 20 mg or placebo throughout the 12-week intervention period.

Patient and public involvement

Patients with lived experience and clinicians with expertise in treating patients with long COVID were involved in the design of this research, including advising on efficacy outcomes, patient burden, recruitment strategies and anticipated concomitant medications in the population.

Participants

As of 15 May 2026, enrolment was completed with a total of 20 clinical sites activated and 203 participants randomised; study completion is anticipated in September 2026. Participants who met all prespecified inclusion criteria (box 1) and none of the exclusion criteria (box 2) were considered eligible. Participants were included if they had fatigue and neurocognitive impairment attributed to an index SARS-CoV-2 infection with symptoms ≥ 3 months as demonstrated by a PROMIS Cognitive Function Short Form (SF)-8a T-score ≤ 40 and a PROMIS Fatigue SF-13a T-score ≥ 50 . Participants were deemed ineligible if they met any of the following criteria, among others (box 2): tested positive for SARS-CoV-2 in the past 28 days; received a vaccination for COVID-19 or influenza within 14 days of randomisation; required admission to the intensive care unit for COVID-19-related symptoms; prior or active unstable or progressive major psychiatric or neurologic conditions; attention deficit/hyperactivity disorder treated with stimulants; fibromyalgia or POTS prior to index infection; currently taking naltrexone at a daily dose > 4.5 mg/day; or received a potent CYP3A4 inhibitor within 14 days prior to randomisation or within five half-lives of the CYP3A4 inhibitor, whichever is shorter. Participants may be withdrawn from the study at the investigator’s discretion or for any of the following reasons: pregnancy, significant protocol

Box 1 Key inclusion criteria

- ⇒ Adult participants ≥ 18 to < 70 years of age at screening.
- ⇒ Long COVID with neurological symptoms as defined below:
 - a. Current symptoms of at least fatigue and neurocognitive impairment that began or worsened after an index SARS-CoV-2 infection that occurred at least 3 months prior to screening. Index SARS-CoV-2 infection is defined as either (1) an episode of COVID-19 with a positive nucleic acid or antigen test during acute illness, as documented in the medical record, or (2) a documented clinical diagnosis of COVID-19, which can be based on a patient-reported positive test for COVID-19.
 - b. Symptoms cannot be explained by any concomitant condition or diagnosis, in the opinion of the investigator.
 - c. Symptom duration of at least 3 months.
- ⇒ PROMIS Cognitive Function SF-8a T-score ≤ 40 (≥ 1 SD worse than normative mean).*
- ⇒ PROMIS Fatigue SF-13a T-score ≥ 50 (worse than normative mean).*
- ⇒ Willing and able to provide consent; participate in the planned study activities; allow the collection of blood for haematology, chemistry, metabolic tests, biomarkers of inflammation and neurodegeneration, DNA methylation analysis, and bezisterim drug level determination; and provide urine for urinalysis and drug screening.
- ⇒ Agree to maintain any other regular medications at current doses for the duration of the trial.
- ⇒ For females taking hormone replacement therapy: the regimen must be stable for ≥ 6 months prior to randomisation and maintained throughout the 12-week treatment period.
- ⇒ Males and females with childbearing potential must agree to an effective method of contraception, such as a barrier method.
- ⇒ Native-level English proficiency.

*PROMIS Fatigue scores are scaled such that higher T-scores indicate greater fatigue severity, whereas PROMIS Cognitive Function scores are scaled such that lower T-scores indicate worse cognitive performance. Thresholds were selected consistent with established normative interpretations. PROMIS, Patient-Reported Outcomes Measurement Information System; SF, Short Form.

violations, if continued participation would not be in the best interest of the participant based on clinical findings or medical conditions, worsening of the disease, if the sponsor terminates the study, if the participant requests to be discontinued at any time, or participation in any other clinical trial. The reason for early discontinuation or withdrawal will be recorded, and participants will be asked to complete an early termination visit. Randomised participants who are withdrawn will not be replaced for any reason.

Study intervention and concomitant therapy

Randomised participants will receive either bezisterim 20mg capsules or matching oral placebo capsules to be taken twice daily. The first dose will be administered and observed on site during the BL visit (visit 2). Aside from the first dose, participants self-administer the study intervention at home and treatment compliance will be assessed via bezisterim blood levels at visits 3, 4 and 5. Participants will be instructed to document the timing of their most recent dose prior to sample collection. Additionally, participants are provided with a treatment diary

Box 2 Key exclusion criteria

- ⇒ Positive SARS-CoV-2 nucleic acid or rapid antigen test in the past 28 days.
- ⇒ Received a vaccination for COVID-19 or influenza within 14 days of randomisation.
- ⇒ Previous admission to the ICU for COVID-19-related symptoms and/or intubated (ie, mechanical ventilation) for COVID-19 care.
- ⇒ Prior or active unstable or progressive major psychiatric or neurologic condition that may impact the ability to determine whether treatment effect is related to SARS-CoV-2 infection, including but not limited to the following:
 - a. Progressive neurodegenerative disease.
 - b. Past traumatic brain injury still associated with active post-concussive symptoms.
 - c. History of epilepsy or seizure disorder requiring ongoing treatment.
 - d. Post-stroke deficits that may interfere with assessments.
 - e. Formal thought disorders.
 - f. Any uncontrolled neuropsychiatric or neurologic disorder.
 - g. Functional neurologic disorder.
 - h. MDD not on stable treatment for at least 3 months prior to screening and not planning to stay on a stable dose through the study, or a PHQ-2 score ≥ 3 . (If the PHQ-2 score is ≥ 3 the investigator should discuss with the Study Medical Monitor to confirm eligibility).
 - i. PMDD.
- ⇒ In the opinion of the investigator, any physical, cognitive (eg, intellectual disability, dementia), or language impairments.
- ⇒ Diagnosed reading disability or dyslexia.
- ⇒ Documented ADHD being treated with psychostimulants. Individuals diagnosed with ADHD being treated with non-stimulants must be stable on the current dosage for ≥ 3 months prior to randomisation and expected to remain stable throughout participation in the study.
- ⇒ Known active bacterial, fungal, viral or other infection besides SARS-CoV-2 requiring treatment within 28 days prior to randomisation and meeting criteria for systemic involvement.
- ⇒ Diagnosis of narcolepsy.
- ⇒ History of obstructive sleep apnoea unless the participant is compliant with prescribed treatment (eg, CPAP or BiPAP therapy) as confirmed by medical records or clinician assessment. History of congestive heart failure, among other cardiac disorders.
- ⇒ Clinically significant ECG findings.
- ⇒ History of moderate or severe COPD or moderate or severe asthma.
- ⇒ History of CFS, fibromyalgia or POTS prior to index COVID-19 infection.
- ⇒ Seropositive for Lyme disease or syphilis.
- ⇒ History of HIV, chronic hepatitis B or hepatitis C.
- ⇒ Screening lab abnormalities that may cause fatigue or cognitive impairment.
- ⇒ Has exclusionary values at screening for ALT, Hb, INR, creatinine clearance or untreated diabetes (HbA1c > 9.0).
- ⇒ Pregnant or lactating.
- ⇒ History of any illness, including cancer, requiring systemic therapy within the last 5 years.
- ⇒ Recent history of thromboembolic events within past 6 months.
- ⇒ History of alcohol use disorder or substance use disorder within 12 months of screening.
- ⇒ History of active suicidal thoughts within 6 months of screening or a suicide attempt within the previous 2 years.
- ⇒ Lifetime history of breast cancer.

Continued

Box 2 Continued

- ⇒ Unwilling to not begin, resume or increase the dose of any form of cognitive training or cognitive-enhancing supplement.
- ⇒ Unwilling to refrain from the use of prohibited medications.
- ⇒ Evidence or suspicion of autoimmune diseases commonly associated with pulmonary involvement and likely to require treatment.
- ⇒ Requirement of supplemental oxygen at the time of screening, unless on a stable supplemental oxygen requirement ≥ 3 months prior to screening.
- ⇒ Received investigational agents within 30 days prior to screening.
- ⇒ Received a direct-acting antiviral agent for COVID-19 infection within the past 2 weeks or TNF inhibitors within the 90 days prior to screening.
- ⇒ Received systemic or intra-articular steroids within 28 days prior to randomisation. An oral corticosteroid regimen of no more than 7 days is allowed within 28 days prior to randomisation. Allowable regimens can be a tapered regimen or, if a tapered regimen is not required, a regimen where single daily doses do not exceed 20 mg/day.
- ⇒ Autoimmune diseases are exclusionary only if the participant is receiving or likely to require initiation of immunomodulatory treatment that would confound the interpretation of the study data during their participation in this study.
- ⇒ Participants taking naltrexone at a dose >4.5 mg/day. Participants on a dose of ≤ 4.5 mg/day must be stable on the current dosage for ≥ 3 months prior to randomisation and expected to remain stable throughout participation in the study.
- ⇒ Use of or requirement for potent CYP3A4 inhibitors — including clarithromycin, erythromycin, diltiazem, itraconazole, ketoconazole, ritonavir or verapamil within 14 days of randomisation or within 5 half-lives of the drug, whichever is shorter, to avoid potential drug interaction through this metabolic pathway.
- ⇒ Any medical intervention, including surgery, planned to occur during the study.
- ⇒ Participants who are relatives of the sponsor or sponsor representatives.

ADHD, attention-deficit/hyperactivity disorder; ALT, alanine aminotransferase; BiPAP, bilevel positive airway pressure; CFS, chronic fatigue syndrome; COPD, chronic obstructive pulmonary disease; CPAP, continuous positive airway pressure; CYP, cytochrome P450; ECG, electrocardiogram; Hb, haemoglobin; HbA1c, glycated Hb; ICU, intensive care unit; INR, international normalised ratio; MDD, major depressive disorder; PHQ-2, Patient Health Questionnaire-2; PMDD, premenstrual dysphoric disorder; POTS, postural orthostatic tachycardia syndrome; TNF, tumour necrosis factor.

to log missed doses and are instructed to bring the diary to visits for verification by the study staff. Dose modification is not anticipated for this study. Any medication or vaccine, including over-the-counter or prescription medicines, recreational drugs, vitamins and/or herbal supplements, that the participant is receiving at the time of enrolment or during the study will be recorded with all relevant information. Participants are permitted to continue allowable pretrial medications during the study if the dose has been stable for ≥ 3 months prior to the screening visit (visit 1) and there are no planned changes in dose to occur during the study.

The following medications will be prohibited throughout the duration of the study: D2 receptor antagonists with exceptions, stimulants, systemic or intra-articular steroids, immunomodulators, vaccinations for COVID-19 or influenza, amantadine and other N-methyl-D-aspartate (NMDA) receptor antagonists and potent CYP3A4 inhibitors. The following restrictions will also apply during the study: sedating medications, alcohol and cannabis-containing products must be withheld for a minimum of 12 hours prior to assessments; nicotine products must be withheld within 4 hours of the visit.

Study endpoints and assessments

This study aims to explore the effect of bezisterim on cognitive function, fatigue, sleep and post-exertional malaise (PEM; [table 1](#)). As a signal-seeking, exploratory phase 2 trial, the primary outcome is intended for estimation and hypothesis generation rather than formal hypothesis testing. As such, all efficacy endpoints will be tested independently without multiplicity adjustment. While change from BL (visit 2) over time will be measured throughout the 12-week treatment period, the change at week 12 (visit 5) will serve as the pre-specified timepoint for all outcomes. To determine the effect of bezisterim on cognitive deficits associated with long COVID, patients will complete the Cogstate Cognition Battery, which will assess 6 areas of cognitive function, including psychomotor function, attention, sustained attention, processing speed, verbal learning and verbal memory. Additional cognitive assessments will include the PROMIS Cognitive Function SF-8a, an 8-item patient-reported questionnaire. Clinical outcomes related to fatigue, sleep and PEM will be assessed through the 13-item PROMIS Fatigue SF-13a score, 8-item PROMIS Sleep Disturbance SF-8a score and the 10-item DePaul Symptom Questionnaire–Post-exertional Malaise score, respectively. To evaluate the quality of life, functional health and well-being, the change in 12-item Short Form Health Survey (SF-12) total and domain scores will be assessed. The effect of bezisterim on the participant's global impression of severity and change will be evaluated through assessing the change in overall and symptom-specific (Fatigue and Think Clearly) Patient's Global Impression of Severity (PGI-S) and Impression of Change (PGI-C). Similarly, the clinician's perspective on global impression of severity and change will be evaluated using the Clinician's Global Impression of Severity (CGI-S) and Impression of Change (CGI-C) questionnaires. The three-item Long COVID Other Burdensome Symptoms Questionnaire will be used to measure the participant's severity and frequency of headache, tinnitus and decreased appetite symptoms. Participants will be asked to report on the occurrence of long COVID fatigue and neurocognitive symptom 'crises', 'crashes' or 'exacerbations' occurring within the 4-week interval prior to the BL visit and during each interval between subsequent visits. The number of reported events and estimated duration and severity (5-point Likert scale) will be recorded. The

Table 1 Objectives and endpoints

Objectives	Endpoints*
Efficacy	
► Evaluate the effect of bezisterim on cognitive function (psychomotor function, attention, sustained attention, verbal learning and processing speed)	► Change in performance on the Cogstate Cognition Battery
► Evaluate the effect of bezisterim on psychomotor function	► Change in Detection test score
► Evaluate the effect of bezisterim on attention	► Change in Identification test score
► Evaluate the effect of bezisterim on sustained attention	► Change in Sustained Attention to Response test score
► Evaluate the effect of bezisterim on processing speed	► Change in Digital Symbol Substitution test score
► Evaluate the effect of bezisterim on verbal learning	► Change in International Shopping List test score
► Evaluate the effect of bezisterim on verbal memory	► Change in International Shopping List Recall test score
► Evaluate the effect of bezisterim on patient-reported perceived cognition deficits	► Change in PROMIS Cognitive Function SF-8a score
► Evaluate the proportion of participants with a clinically meaningful improvement in cognition	► Proportion of participants with an improvement of ≥ 0.5 SD in the Cogstate Cognition Battery score
► Evaluate the effect of bezisterim on fatigue	► Change in PROMIS Fatigue SF-13a score
► Evaluate the effect of bezisterim on sleep	► Change in PROMIS Sleep Disturbance SF-8a score
► Evaluate the effect of bezisterim on post-exertional malaise	► Change in DSQ-PEM score
► Evaluate the effect of bezisterim on quality of life, functional health and well-being	► Change in the SF-12 total and domain scores
► Evaluate the effect of bezisterim on participant's global impression of severity and change in symptoms	► Change in overall and symptom-specific (Fatigue and Think Clearly) PGI-S and PGI-C
► Evaluate the effect of bezisterim on participant's impression of severity and frequency of other long COVID symptoms	► Change in participant's perception of severity and frequency of other burdensome long COVID symptoms (headache, tinnitus and decreased appetite)
► Evaluate the effect of bezisterim on participant's reported incidence, severity and duration of long COVID fatigue and neurocognitive symptom exacerbations	► Change in participant's reported incidence, severity and duration of symptoms on the long COVID fatigue and neurocognitive symptom exacerbation
► Evaluate the effect of bezisterim on clinician's global impression of severity and change in symptoms	► Change in CGI-S and CGI-C
Exploratory	
► Assess the effect of bezisterim on inflammatory biomarkers	► Change in plasma biomarkers of inflammation/immunity over 12 weeks
► Assess the effect of bezisterim on biomarkers related to neurodegeneration and metabolic dysfunction	► Change in plasma biomarkers of neurodegeneration and metabolic dysfunction over 12 weeks
► Assess the effect of bezisterim on biological age	► Change in DNA methylation over 12 weeks
Safety	
► Assess the safety and tolerability of bezisterim	► Incidence and severity of TEAEs ► Incidence of clinically significant changes in vital signs (blood pressure, pulse, respiratory rate and temperature) ► Incidence of clinically significant changes in physical examinations ► Incidence of clinically significant changes in 12-lead ECGs ► Incidence of clinically significant changes in clinical laboratory assessments (haematology, chemistry and urinalysis)

*Evaluated at V5 (EOT; week 12).

CGI-C, Clinician's Global Impression of Change; CGI-S, Clinician's Global Impression of Severity; DSQ-PEM, DePaul Symptom Questionnaire-Post-exertional Malaise; ECG, electrocardiogram; EOT, end of treatment; PGI-C, Patient's Global Impression of Change; PGI-S, Patient's Global Impression of Severity; PROMIS, Patient-Reported Outcomes Measurement Information System; SF, Short Form; SF-12, 12-Item SF Health Survey; TEAE, treatment-emergent adverse event; V, visit.

PHQ-2, Epworth Sleepiness Scale and C-SSRS Baseline/Screening will be administered at visit 1 to determine eligibility criteria. Following the initial screening period (visit 1), the C-SSRS since last visit will be administered at visits 2–5 for continued monitoring.

Exploratory endpoints include the effects of bezisterim on inflammatory biomarkers and other biomarkers related to neurodegeneration and metabolic dysfunction as assessed by blood and plasma samples collected throughout the study period. Blood samples for DNA methylation and RNA expression analyses, and plasma samples for biomarkers of inflammation and neurodegeneration, will be collected and assessed at each visit during the intervention period to enhance the understanding of the biological underpinnings of neurological symptoms associated with long COVID. Blood samples for bezisterim drug level determination will be collected and assessed at visits 3–5 (online supplemental exhibit 1; online supplemental table S1). Participants are made fully aware of the number, purpose and timing of blood collections, which are minimal in sampling volume and frequency while still allowing for assessment of pharmacokinetics (PK) and exploratory objectives. Additionally, sampling is aligned with scheduled study visits to avoid additional visit burden.

The incidence and severity of treatment-emergent adverse events (TEAEs) will be identified through safety assessments (table 1) that will be performed during scheduled visits (online supplemental exhibit 1). Any clinically significant changes occurring during the study, including clinically significant changes in vital signs, physical examinations, electrocardiograms and clinical laboratory assessments, will be recorded as an adverse event (AE). The clinical laboratory tests will include assessments of haematology, chemistry and urinalysis (table 2). All clinical laboratory tests with values considered significantly abnormal during the study will be repeated until the values are no longer considered clinically significant by the investigator. The study may be discontinued at the discretion of the sponsor or the sponsor's medical monitor/investigator based on increased frequency, severity and/or duration of known AEs or any other concerns regarding safety.

Blood samples will be collected to evaluate trough levels of bezisterim and dosing compliance, with instructions to document timing of the most recent dose prior to sample collection. Whole blood will be analysed for DNA methylation and RNA expression and then stored frozen; no gene sequencing will be conducted. Plasma samples will assess levels of biomarkers of inflammation,

Table 2 Clinical laboratory tests

Haematology	RBC count, Hb, HCT, platelet count (absolute), WBC count with differential, coagulation (INR, PT and PTT)
Chemistry	Sodium, potassium, chloride, GGT, ALT, AST, albumin, ALP, urea, calcium,* creatinine, total bilirubin, hs-CRP, blood glucose, amylase, lipase, HDL, LDL, LDH, triglycerides, VLDL, total cholesterol
Metabolic	HbA1c
Urinalysis	pH, specific gravity, protein, glucose, ketones, bilirubin, nitrite, urobilinogen, leucocyte esterase [†]
UDS	A commercially available kit will be used to screen for amphetamines, barbiturates, benzodiazepines, cocaine and metabolites, methadone, methamphetamines, MDMA, opiates, oxycodone, phencyclidine, and cannabinoids (THC) [‡]
Pregnancy	hCG [§]
FSH*	FSH (post-menopausal women only)
Thyroid*	TSH, Free T4
Autoimmunity	ANA

*Assessed at screening only.

[†]If urinalysis is positive for protein, blood, nitrite and/or leucocyte esterase, a microscopic examination (for RBCs, WBCs, bacteria, casts and epithelial cells) will be performed.

[‡]If screening UDS reports only the presence of THC, confirmatory testing by the central lab will not be required.

[§]Childbearing potential is defined as a premenopausal female capable of becoming pregnant. This includes women using oral, injectable or mechanical contraception; women who are single; and women whose husbands have been vasectomised or whose husbands have received or are using mechanical contraceptive devices. Women who have been surgically sterile for at least 6 months or who are postmenopausal (no menstrual bleeding for at least 1 year) will not be considered to be of childbearing potential. A serum pregnancy test will be performed on participants of childbearing potential at screening. A urine/dipstick pregnancy test will be performed at all subsequent visits; participants who test positive will be referred for care as per state and institutional practice guidelines.

^{||}Evaluated at V2 (BL; day 1) and V5 (EOT; week 12) only.

ALP, alkaline phosphatase; ALT, alanine aminotransferase; ANA, antinuclear antibody; AST, aspartate aminotransferase; BL, baseline; EOT, end of treatment; FSH, follicle-stimulating hormone; GGT, gamma-glutamyl transpeptidase; Hb, haemoglobin; HbA1c, glycated Hb; hCG, human chorionic gonadotropin; HCT, haematocrit; HDL, high-density lipoprotein; hs-CRP, high-sensitivity C-reactive protein; INR, international normalised ratio; LDH, lactate dehydrogenase; LDL, low-density lipoprotein; MDMA, 3,4-methylenedioxymethamphetamine; PT, prothrombin time; PTT, partial thromboplastin time; RBC, red blood cell; THC, tetrahydrocannabinol; TSH, thyroid-stimulating hormone; UDS, urine drug screen; V, visit; VLDL, very low-density lipoprotein; WBC, white blood cell.

which may include C1q, interferon-gamma (IFN- γ), IL-1 β , IL-6, IL-17, IL-23, TNF, cytokine regulated on activation, normal T-cell expressed and secreted, and monocyte chemoattractant protein-1 and neuroinflammation/neurodegeneration (phospho-tau, neurofilament light, glial fibrillar acidic proteins, amyloid beta42/beta40). Antinuclear antibodies will be collected at visit 2 and visit 5 only. If multiple assessments requiring blood samples and safety laboratory assessments are to be evaluated at the same time, a single blood sample may be collected and aliquoted as appropriate to allow the specific analyses to be performed (online supplemental exhibit 1; online supplemental table S1).

Data will be analysed as specified in the statistical analysis plan. The analysis populations will include the following: (1) the modified intention-to-treat (mITT) population, defined as all participants who received at least one dose of the study intervention and had at least one post-BL efficacy measurement; (2) the per-protocol population, defined as all participants in the mITT population who had no important protocol deviations affecting efficacy and safety assessments; and (3) the safety population, defined as all participants who received at least 1 dose of the study intervention. Any non-compliance with the clinical trial protocol and associated documents on the part of the participant, the investigator, and/or the study site staff is considered a protocol deviation. Corrective actions for deviations are to be developed by the site and implemented promptly. The mITT population will be used for analysis of the safety measures, unless otherwise specified. BL information will include measurements of demographic and clinical characteristics of participants collected before the first administration of study intervention, and on-treatment (ie, longitudinal) data are collected thereafter until discharge and 4-week follow-up. The primary analysis model to assess change from BL at week 12 for all efficacy endpoints will be a mixed model for repeated measures using restricted maximum likelihood in the mITT population with treatment, visit and treatment-by-visit as fixed effects; BL score as a covariate; and participants as a random effect. The secondary analysis for the efficacy endpoints will be an analysis of covariance, where missing endpoints will be imputed with the last (post-BL) observation carried forward. Sensitivity analysis to explain the impact of different assumptions on missingness will be conducted, that is, missing at random and missing not at random.

Summary statistics will be provided for TEAEs, serious AEs (SAEs) and study discontinuation due to AEs of death. Counts and proportions of participants experiencing AEs will be reported for each treatment group.

Exploratory analyses will be conducted to investigate potential correlations between clinical outcomes and a range of biological markers. These analyses aim to generate hypotheses and provide mechanistic insights that may inform future studies and therapeutic strategies. Specifically, we intend to support identified

candidate biomarkers for future validation by evaluating the following:

- Correlations between BL biomarker levels and treatment response, including inflammatory cytokines (eg, IL-6, TNF- α , IL-1 β), neurodegenerative markers (eg, phospho-tau, amyloid beta42/40), and metabolic indicators.
- Longitudinal changes in biomarker profiles and their associations with symptom improvement, as measured by validated clinical scales and patient-reported outcomes.
- Biomarker signatures predictive of responder status, using stratified analyses based on symptom duration, severity, and cognitive function.
- Exploratory subgroup analyses to assess whether specific biomarker patterns are associated with differential treatment effects across demographic or clinical subgroups.
- Multivariate modelling to identify potential interactions between biomarkers and clinical endpoints, adjusting for relevant covariates.

Sample size considerations

As a proof-of-concept study, efficacy endpoints will be evaluated independently, and no multiplicity adjustment has been applied in the sample size determination. Assuming approximately 81 participants per treatment arm complete the week 12 assessment, the study is powered at 80% (two-sided $\alpha=0.05$) to detect a between-group difference of approximately 0.4 SD units in the mean change from BL to week 12 in Cogstate domain scores or PROMIS measures. For PROMIS instruments, this corresponds to approximately 4 T-score points (SD=10). The sample size assumptions are based on an anticipated SD of approximately 9; notably, in RECOVER-NEURO, the SD for mean change in the PROMIS Cognitive Function SF-8a score in the active comparator arm was estimated at 7.4 (derived from a 95% CI; n=61), supporting the plausibility of this assumption.⁴⁸

At present, there is no universally established minimally important difference (MID) for PROMIS measures in long COVID populations. However, a difference of 0.4 SD (approximately 4 T-score points) is consistent with published MID ranges for PROMIS domains in other clinical contexts and represents a small-to-moderate treatment effect. This threshold is also broadly aligned with assumptions used in contemporaneous long COVID trials; for example, the RECOVER-SLEEP protocol defines a clinically meaningful treatment effect as a 5-point change in PROMIS T-scores.⁴⁹

Under these assumptions, the study is expected to have sufficient sensitivity to detect clinically meaningful directional signals between bezisterim and placebo over 12 weeks of treatment, although the trial is exploratory and not designed for confirmatory hypothesis testing.

Data generation, analysis, management and retention

Clinical data, including AEs and concomitant medications, and other data (eg, clinical laboratory data) will be entered into an electronic data capture system that complies with regional and national regulatory standards and is 21 CFR Part 11 compliant. The data system will require password protection, and internal quality checks will be performed to identify inconsistencies. Sites will be trained to maintain accuracy, completeness, legibility and timeliness of reported data. A data management plan will also be prepared, including detailed data management activities and quality assurance. Investigators will maintain confidentiality for all study documentation and will take measures to prevent accidental or premature destruction of these documents. The investigators, investigational sites and institutional review board (IRB)/independent ethics committee agree to retain study-related documents in a secure location for at least 2 years after the last approval of a marketing application following formal discontinuation of the investigational product.

Ethics and dissemination

The protocol, informed consent form (ICF), investigator's brochure and other relevant documents (eg, advertisements) will be submitted to an IRB by the investigator to be reviewed and approved before the initiation of the study. The investigator will provide written summaries of the status of the study to the IRB at least annually in accordance with the requirements, policies and procedures established. Additionally, the investigator will notify the IRB of SAEs and other significant safety findings, as required. The study protocol has received central IRB approval (Advarra #Pro00081088).

This study will be conducted in accordance with the protocol and the consensus ethical principle as derived from the Declaration of Helsinki and Council for International Organizations of Medical Sciences International Ethical Guidelines, applicable International Council for Harmonization (ICH) Good Clinical Practice (GCP) and applicable laws and regulations.

A clinical research organisation has been contracted by the sponsor to oversee the conduct of the trial per GCP regulations and thus is not independent of the sponsor. Likewise, the statistical analysis plan, database access and programming are being managed by a statistical consulting company contracted by the sponsor. Consent forms describing the study intervention, study procedures and risks will be given to each potential participant to provide an explanation of the nature and purpose of the study. Potential participants will have the opportunity to review the written ICF, discuss with their families and ask questions as needed prior to signing the document. Written documentation of informed consent (paper or electronic) will be obtained from each participant prior to starting the study. Strict participant confidentiality and privacy will be observed by the participating investigators, their staff and the sponsor. Biological samples, genetic testing and clinical information related to participants

will be confidential and will not be released to any unauthorised third party without prior written approval from the sponsor. The site monitor, authorised representatives of the sponsor and representatives of the IRB or regulatory agency may inspect all relevant documents and records that are required to be maintained by the investigator, including medical and pharmacy records.

The sponsor will maintain confidentiality standards by assigning a unique study-specific number to each participant enrolled in the study. The contact information for study participants will be securely stored at each clinical site for internal use during the study. Data generated by this study will be available upon request by representatives of national and local health authorities, the sponsor (or its designee) and the IRB for each study site, as appropriate. Each clinical site will be responsible for internal quality management of study conduct, data, biological specimen collection, documentation and completion based on its internal processes. Following written standard operating procedures and the clinical monitoring plan, site monitors will verify that biological specimens are collected in compliance with the protocol, ICH GCP and applicable regulatory requirements (eg, Good Laboratory Practices and Good Manufacturing Practices).

Data collected from participants are intended for use only by the study sponsor, investigators and designated study collaborators for the purposes of conducting and evaluating the clinical trial and related regulatory submissions. Following peer-review/publication of the primary study results, access to the aggregate data, study protocol and statistical analysis plan will be considered for methodologically sound proposals upon reasonable request. Data sharing will be conducted in a manner that protects participant privacy, maintains compliance with applicable regulations and safeguards proprietary and confidential information. Data will be shared with lead study investigators, patient advisors and the US Food and Drug Administration. Final interpretation of the data will be made in collaboration with the sponsor under the direction of the Chief Medical Officer.

DISCUSSION

There is an urgent unmet need for treatments that reduce the burden of cognitive impairment and fatigue in long COVID. Despite intensive global research efforts, multiple interventional studies have failed to demonstrate clinical benefit in long COVID, including the much anticipated National Institutes of Health-sponsored interventional study RECOVER-NEURO, which did not meet its primary endpoint, modified Everyday Cognition Scale 2.^{48 50 51} These findings underscore the complexity of long COVID pathophysiology and the challenges inherent in demonstrating treatment effects in heterogeneous populations using predominantly PROs.

Previous studies have shown the activity of low-dose naltrexone (LDN) on a wide range of off-label uses for inflammation and pain, specifically through its activity

against non-opioid receptors (eg, TLR4 receptors). In a cohort study, Bonilla *et al* indicated that participants who received LDN off-label for PASC self-reported improvements in fatigue, sleep patterns and functional status believed to be due to its anti-inflammatory and immunomodulatory effects, possibly warranting clinical trials.⁵² A phase 2, randomised, double-blind, placebo-controlled study has completed recruiting participants to determine if LDN reduces fatigue, inflammatory markers and symptoms related to PASC (NCT05430152).⁵³

Bezisterim has been previously evaluated for cognitive conditions, such as PD and AD, and was considered well tolerated with no SAEs associated with the study treatment.⁴² Our current understanding of long COVID pathophysiology suggests the need for a therapy that targets multiple mechanisms of disease. Due to its mode of action, bezisterim may decrease the need for multiple medications, potentially providing an alternative to other currently investigated therapies that may only address one underlying mechanism of disease, such as reducing viral reservoir.

This is the first study evaluating the impact of bezisterim on cognitive and neurological symptoms associated with long COVID. The randomised, triple-blind, placebo-controlled study design minimises bias and expectation effects, which is particularly important in trials that include subjective outcomes. Additionally, to increase the likelihood of signal detection, participant selection is enriched for subjective cognitive dysfunction and subjective fatigue.

Ideally, clinical trials in long COVID should incorporate a decentralised or hybrid design to minimise participant burden, as the condition is often associated with severe fatigue, cognitive dysfunction and limited functional capacity that can make travel to study sites challenging. Decentralised approaches have the advantage of reducing logistical barriers and improving access and participation by allowing study activities to be conducted in the patient's home environment. However, for a signal-seeking phase 2 study in which efficacy is primarily assessed through PROs, the reliability, consistency and interpretability of these subjective measures are critical. As PROs are inherently dependent on patient perception, they are sensitive to variability in administration conditions, timing and understanding of the measurement tool. Accordingly, administration of key PRO assessments in a controlled, site-based environment was prioritised in this study to reduce variability, ensure standardised instructions and optimise data quality for early efficacy assessment. While on-site visits may impose a burden on participants with long COVID, the protocol was designed to minimise this burden by limiting study procedures to those essential for interpretation of treatment effect. Specifically, PRO assessments were restricted to a core set of clinically meaningful instruments, and blood sampling and ancillary evaluations were limited to timepoints and measures necessary to support interpretation of clinical and mechanistic outcomes. This design reflects a deliberate balance

between minimising participant burden and preserving the methodological rigour required to detect a meaningful signal in a heterogeneous and poorly characterised condition.

Several limitations of the trial design should be considered. The absence of a validated biomarker for long COVID introduces diagnostic uncertainty and potential heterogeneity, which may dilute treatment effects. Reliance on PROs to assess brain fog, fatigue and PEM increases susceptibility to placebo response and reporting variability. Additionally, the sample size and duration typical of phase 2 studies may limit the detection of smaller or longer-term treatment effects.

In summary, this proof-of-concept study was designed to rigorously evaluate the safety and efficacy of bezisterim in long COVID. The design seeks to balance methodological rigour with the practical challenges inherent in this population. Findings are expected to inform future confirmatory studies and contribute to the development of treatments for long COVID.

Author affiliations

¹BioVie, Carson City, Nevada, USA

²University of California San Francisco, San Francisco, California, USA

³Feinberg School of Medicine, Northwestern University, Evanston, Illinois, USA

⁴Patient-Led Research Collaborative, Oakland, California, USA

⁵Perissos, Wake Forest, North Carolina, USA

⁶Biotechant Solutions LLC, Newtown, Pennsylvania, USA

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Competing interests PM, CR, CA and JY are employees and shareholders of BioVie Inc. MJP, LM, SOQ and CMP have received consulting fees from BioVie Inc. SH-YC has received consulting fees from Grace Therapeutics. CMP is a consultant and an employee of Biotechant Solutions LLC. JMP is an employee of BioVie Inc., is a shareholder, has employee stock options, and has the following patents approved or pending: BVIE.020W0, BVIE.028PR. Unrelated to this submitted work, MJP has received consulting fees from Gilead Sciences, AstraZeneca, Apellis Pharmaceuticals and BioNTech, and research support from Aerium Therapeutics and Shionogi.

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ORCID iD

Penelope Markham <https://orcid.org/0000-0001-6412-676X>

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